

BT₅₀₄ (Genomics and Proteomics)

Midterm – Spring 2023 (Current + Past)

Solved By Bella

We Grow Together

Write down main format of microarray technique? (3)

- Cartridge-based Spotted Electronic
- Spotted Glass Slide
- Tissue Section Slide

Write different techniques of physical mapping?

- Cytogenetic map
- A restriction map
- Fluorescent in situ hybridization
- Sequence tagged site (STS) map
- Nucleotide sequence map

What is prerequisite is determination of genome sequence?

Determination of entire genome sequence is a prerequisite to understand the complete biology of an organism.

Variation in bacterial genome?

- *Tremblaya princeps* 140kb, 121 coding sequences
- *Sorangium cellulosum*
- 14000kb
- 11599 coding sequences

Position independent method?

- No information about the location of the gene.
- Starting at the phenotype, determining which protein was involved and getting to gene through the protein.
- Starting from the approximate location of gene, to finding the gene itself, then translating it to learn about the protein and its function.
- Positional cloning or reverse genetics
- Mostly identifying and isolating the protein product of the gene.
- Such genes usually produce large amounts of well-known and well studied proteins.
- Two common methods used
- Gene-specific oligonucleotides
- Antibody Methods

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- Recombination mapping and/or somatic cell mapping to defined the region of interest as tightly as possible.
- Chromosomes are inherited as intact units, so it is reasoned that the alleles of some pairs of genes will inherit together because they are on the same chromosome.
- Different approaches are used for identification of disease genes

3 Function of miRNA?

1. Involved in the post-transcriptional regulation of gene expression
2. Important in development
3. Metabolic regulation (miR-375 & insulin secretion)

Insertion sequence elements?

- The simplest transposable elements
- Code only for the ability to transpose
- Prokaryotes
- Usually very small (< 1 Kb to 2 Kb)
- Flanked by inverted repeat sequences (IRs)
- Encode at least one gene that provides their own transposition functions
- Do not code for noticeable (phenotypic) traits
- Can cause mutations by transposition into genes

Difference between eukaryotic and prokaryotic DNA transcription?

Prokaryotes

- Operons
- 27% of E. coli genes (Housekeeping genes not in operons)
- Simultaneous transcription and translation

Eukaryotes

- No operons, but they still need to coordinate regulation.
- More kinds of control elements
- RNA processing
- Chromatin remodeling.
- Histones must be modified to loosen DNA.

Why Microarray needs to be scan, different colors and their result.

- The scanner has a laser, a computer, and a camera.
- The laser causes the hybrid bonds to fluoresce.
- The camera records the images produced when the laser scans the plate.
- The computer allows to immediately view the results and it also stores the data.
- GREEN – the healthy sample hybridized more than the diseased sample.
- RED – the diseased/cancerous sample hybridized more than the nondiseased sample.
- YELLOW – both samples hybridized equally to the target DNA.
- BLACK – areas where neither sample hybridized to the target DNA.

Steps of genome analysis?

- Genome sequence assembled
- Identify repetitive sequences – mask out
- Gene prediction – train a model for each genome.
- Look for EST and cDNA sequences
- Genome annotation
- Microarray analysis
- Metabolic pathways and regulation
- Protein 2D gel electrophoresis.
- Functional genomics
- Gene location/gene map
- Self-comparison of proteome
- Comparative genomics
- Identify clusters of functionally related genes
- Evolutionary modeling

D-loop of human mitochondrial genome? (5)

- Circular, double stranded, 16.6 kb
- The two strands are notably different in base composition, leading to one strand being heavy (H strand) and the other light (L strand).
- Both strands encode genes, although more are on the H strand.
- A short region (1121 bp), the D loop is a DNA triple helix: two overlapping copies of the H strand.
- The D loop is also the site where most of replication and transcription is controlled.
- Genes are tightly packed, with almost no non-coding DNA outside of D loop.
- Human mitochondrial genes contain no introns, although introns are found in the mitochondria of other groups (plants)
- 37 Genes
- 22 tRNAs
- 2 rRNAs
- 13 polypeptides
- tRNA: only 60 of the 64 codons code for amino acids.

Advantage of microarray? (5)

- Provides data for thousands of genes
- One experiment instead of many
- Fast and easy to obtain results
- Closer to discovering cures for diseases and cancer
- Different parts of DNA can be used to study gene expression

Define physical mapping? 3

Physical mapping is based on the direct analysis of DNA. Physical mapping places genes on the genomes in relation to distances measured in bp, kbp, and mbp.

What are the benefits of genomics (3)

- Clinical diagnostics and predictive testing
- Identifying new treatment
- Developing preventive measures
- Direct economic benefits
- The human genome is the basis for developing a variety of products to improve the identification, treatment and prevention of ill health.
- Identification of disease genes can have applications in clinical diagnostics and predictive testing. For example, the polymerase chain reaction (PCR) technique, a highly sensitive and accurate genome-based diagnostic method that multiplies the DNA of a pathogen making it easier to detect, can be simplified for use in poor countries to diagnose infectious diseases, such as leishmaniasis and dengue fever, more rapidly, accurately and at less cost.
- New treatment can be developed by identifying new drug targets or tailoring drugs to specific genetic characteristics of individuals. Eg differential drug absorption can influence dosing levels and side effects.
- In developed preventive measures, genomics can, for example, help identify multiple vaccine targets, useful in developing vaccines for organisms with a complex multi-host life cycle, such as the malaria parasite.
- Genomics will also be a significant contributor to the biotechnology sector, which has major income generating potential, expected to grow from US\$2.2 billion in 1999 to US\$8.2 billion in 2004. This will not only benefit developed countries. Eg Cuba, investing heavily since the 1980's in biotechnology, produces several successful products, including the world's only meningitis B vaccine and holding at least 400 patents in the biotech field.

Write the steps of hierarchical shotgun sequencing? (5)

HGP – Hierarchical Shotgun Sequencing

- Map Construction
- Clone selection
- Sub-clone library construction
- Random shotgun phase
- Directed finishing phase and sequence authentication

Hierarchical Shotgun Sequencing: Map Construction

- Clone genomic DNA in YACs (~1MB) or BACs (~200-300Kb)
- Map the relative location of clones
- Sequenced-tagged sites (STS, e.g. EST) mapping
- PCR or probe hybridization to screen STS
- Restriction site fingerprint
- Most time consuming
- 1990-98 to generate physical maps for human

What is microarray? Also write its synonyms, principle, uses and its advantages. (5)

Microarray

Microarray technology evolved from Southern blotting, where fragmented DNA is attached to a substrate and then probed with a known DNA sequence.

Synonyms of Microarray

There are several synonyms of DNA microarrays such as

- DNA chips
- Gene chips
- DNA arrays
- Gene arrays
- Biochips

Microarray – Principle

The principle of DNA microarrays lies on the hybridization between the nucleotides.

Microarray Use

- Determine which genes are active in a cell and at what levels
- Compare the gene expression profiles of a control vs treated
- Assessing gene expression levels
- Genome-wide studies and genotyping
- Evaluating microRNA levels
- Determine which genes have biological significance in a system
- Discovery of new genes, pathways, and cellular trafficking

Microarray – Advantages

- Provides data for thousands of genes.
- One experiment instead of many.
- Fast and easy to obtain results.
- Closer to discovering cures for diseases and cancer.
- Different parts of DNA can be used to study gene expression.

Biogenesis of miRNA? 3 Marks

- Primary-miRNA is transcribed in the nucleus and is usually several kilo-bases long having 5' cap and a poly-A tail.
- Cleaved in the nucleus by Drosha enzyme to 70nt hairpin transcript (pre-miRNA).
- Transported to the cytoplasm by Exportin 5 through nuclear pores.
- Cleaved by Dicer enzyme (RNase II enzyme) into 19-22nt ds-transcripts.

Origin of miRNA?

- miRNA originates from ssRNA that forms a hairpin secondary structure
- miRNA regulates post-transcriptional gene expression and is often not 100% complementary to the target
- Originate from capped & polyadenylated full length precursors (pri-miRNA)
- Hairpin precursor ~70 nt (pre-miRNA)
- Mature miRNA ~22 nt (miRNA)

Name the building blocks of gene rich and gene poor area?

- Gene poor areas are rich in building blocks A and T.
- GC- and AT-rich regions can be seen through microscope as light and dark bands-chromosomes.
- Genes appear to be concentrated in random areas along the genome.
- Large areas of noncoding DNA between the genes.

What is the genome size of bacteria and archaea?

Free-living bacteria and archaea have 1,500 to 7,500 genes.

Describe D loop in human mitochondrial genome? 3 Marks

Human Mitochondrial Genome

- The D loop is also the site where most of replication and transcription is controlled.
- Genes are tightly packed, with almost no non-coding DNA outside of D loop.
- Human mitochondrial genes contain no introns, although introns are found in the mitochondria of other groups (plants).

Define homologue and its subdivisions? 5 Marks

Homologues

Homologues are thus components or characters (such as genes/proteins with similar sequences) that can be attributed to a common ancestor of the two organisms during evolution.

Homologous can be

1. Orthologous

Orthologues are homologues that have evolved from a common ancestral gene by speciation. They usually have similar functions.

2. Paralogues

Paralogues are homologues that are related or produced by duplication within a genome followed by subsequent divergence. They often have different functions.

3. Xenologues

Xenologues are homologous that are related by an interspecies (horizontal transfer) of the genetic material for one of the homologues. The functions of the xenologues are quite often similar.

4. Analogues

Analogues are non-homologues genes/proteins that have descended convergently from an unrelated ancestor. Similar function, different sequence or structure.

Pseudogenes? 3 marks

- Non functional genes
- Derived from functional genes through mutations following genome duplication
- Processed pseudogenes lack introns and promoter.

Super Families? 3 marks

- Super families are much larger than single multigene families.
- Super families contain hundreds of genes, including multiple multigene families and single individual genes

Limitations of Restriction Mapping? 3 marks

Restriction mapping is more applicable to small rather than large molecules. In practice, if a DNA molecule is less than 50 kb in length it is usually possible to construct a restriction map. To construct a map for large genomic size is difficult.

What are the limitations of the microarray?

- Cross-hybridization of sequences with high identity
- Chip to chip variation
- The real limitation is Bioinformatics
- Expensive – repeat experiments

What is gene Discovery and Disease Diagnostics?

Classify the types of cancer on the basis of the patterns of gene activity in the tumor cells.

Define Toxicogenomics and Pharmacogenomics?

Toxicogenomics

Microarray technology allows us to research the impact of toxins on cells. Some toxins can change the genetic profiles of cells.

Pharmacogenomics

Study of correlations between therapeutic responses to drugs and the genetic profiles of the patients.

What are the Biological Applications of microarray?

- Many biological discovery
- New and better molecular diagnostics
- New molecular targets for therapy
- Finding and refining biological pathways

What is CpG? 3 marks

Mouse genome contains fewer CpG islands (15,500). The (G+C) content for each of the mouse chromosome is relatively similar.

Most human genes have mouse orthologs

- Coding exons usually correspond 1-1
- Coding sequence similarity ~ 85%

Explain human genome project? 5 marks

Human Genome Project

- 1990 – Human Genome Project was started (NHGRI)
- Later many institutes of UK, France, Japan, Germany, China involved.
- In 1998, Celera announced a 3-year plan to sequence human genome

Challenges to Sequence Human Genome

- Size
- Polymorphism
- Repeats – smaller repeats are technically difficult to sequence
- Some DNA sequence are repeated all over the genome.
- Relies on cloning (Some regions are difficult to clone - Heterochromatin).
- Some sequences rearrange or are deleted when cloned.

Human Genome Project – Objectives

- Create a genetic and physical map of the 24 human chromosomes (22 autosomes, X & Y)
- Identify the entire set of genes & map them all to their chromosomes.
- Determine the nucleotide sequence of the estimated 3 billion base pairs
- Analyze genetic variations in humans
- Map and sequence the genomes of model organisms
- Two strategies used to sequence human genome
- Hierarchical Shotgun Sequencing
- Whole Genome Shotgun Sequencing

What are the noncoding sequence of the genome?

Non-coding DNA. Noncoding DNA sequences are components of an organism's DNA that do not encode protein sequences. Some noncoding DNA is transcribed into functional non-coding RNA molecules (e.g. transfer RNA, ribosomal RNA, and regulatory RNAs).

Write the basic criteria of DNA sequence?

Following are the criteria for organisms selected for sequencing;

- They had been subjected to a detailed biological analysis and thus were model organisms
- Important human pathogens
- They were of phylogenetic interest
- Sequences were annotated as they were sequenced

Difference between sequencing of eukaryotic and prokaryotic genome? 5 Marks

Prokaryotic Genome

- Prokaryotes are the organisms whose Genetic material (DNA) is not enclosed in a nuclear membrane
- No membrane bound organelles
- First prokaryotic Genome sequenced was that of Hemophilus influenza.
- Paved the way for sequencing of many other organisms.

Eukaryotic Genome

- Eukaryotes have larger genomes
- Have tandem repeats
- Have introns in their protein-coding genes
- Heterochromatin and euchromatin region
- Dense heterochromatin
- Light Euchromatin
- Heterochromatic regions are poorly transcribed (expressed)
- Euchromatic regions are highly expressed

Write down the types of protein with one example?

Different method of categorizing proteins but there are three major categories of proteins.

Fibrous Proteins – Cytoskeletal Proteins

- Actin
- Coronin
- Dystrophin
- Keratin
- Tubulin

Fibrous Proteins – Extracellular Matrix Proteins

- Collagen
- Elastin
- Fibronectin

Globular Proteins - Major Types

- Plasma proteins
- Hemoproteins
- Cell adhesion
- Transmembrane transport proteins.
- Hormones and growth factors
- Receptors
- DNA-binding protein
- Immune system proteins
- Nutrient storage/transport
- Chaperone proteins
- Enzymes

Complexes with multiple components including proteins

- Nucleosome
- Ribonucleoprotein (generic)
- Signal recognition particle
- Spliceosome

Advantages of sequenced genome (like which is already sequenced)? 5

When looking at a whole genome sequence, health professionals consider a number of variables to determine your risk of disease or the best course of treatment for a disease. Some include: Mutations in single genes which can cause single-gene disorders, like muscular dystrophy. When looking at a whole genome sequence, health professionals consider a number of variables to determine your risk of disease or the best course of treatment for a disease. Some include:

- Mutations in single genes which can cause single-gene disorders, like muscular dystrophy.
- DNA changes in multiple genes which can work in tandem to create a disease. An example of one of these multi-factorial disorders is diabetes.
- Pharmacogenomics profile which uses a person's genetic map to figure out how a person will respond to a medication or certain treatment.
- Armed with the findings from whole genome sequencing, you and your physician now have the ability to make smarter health-related decisions and prepare your family for the best healthcare.

Define Intron? 2 marks

They are non-coding sections of a gene. Transcribed in the precursor mRNA sequence but is ultimately removed by RNA splicing. Most of the introns appear to be mobile genetic elements.

What is Yeast genome and its characteristics? 5 Marks

- The nuclear genome consists of 16 chromosomes.
- In addition, there is a mitochondrial genome and a plasmid, 2 micron circle.
- The haploid yeast genome consists of ~ 12.1 Mb
- Yeast genome was completely sequenced by 1996.
- Small and compact
- Small intergenic sequences
- Few transposable elements
- Few introns
- Limited RNA interference
- The yeast genome is predicted to contain about 6,200 genes
- 274 tRNA
- 287 introns
- Small percentage of yeast genes have introns.
- The intergenic space between genes is only between 200bp – 1,000bp

Describe Mouse Genome Contigs?

- Most mouse BAC contigs contained multiple mouse markers.
- Coverage of the mouse genome (2.6 billion bp) in mapped BACs is virtually complete: 296 contigs of average size 9.3 Mb
- Alignment between part of human chromosome 6 and mouse chromosome 4 . 1.6-Mb interval is enlarged, showing part of 16.1 aligned to a 1.3-Mb mouse BAC contig. 11 of the 15 segments of human sequence match to 29 of segments of mouse BAC contig

Process of Protein translation? 5 marks

Protein translation is the process by which a ribosome reads the genetic code in mRNA and synthesizes a protein. The simple steps of protein translation include:

1. **Initiation**: The ribosome binds to the mRNA at the start codon, and a tRNA carrying the amino acid methionine binds to the ribosome.
2. **Elongation**: The ribosome moves along the mRNA, reading each codon and bringing in the appropriate tRNA molecule with the corresponding amino acid. Peptide bonds are formed between the amino acids to build the growing protein chain.
3. **Termination**: When the ribosome reaches a stop codon on the mRNA, protein synthesis stops, and the protein is released from the ribosome, folding into its functional shape.

Define CDNA?

In genetics, complementary DNA is DNA synthesized from a single-stranded RNA template in a reaction catalyzed by the enzyme reverse transcriptase.

D- Loop? 3 Marks

The D loop is also the site where most of replication and transcription is controlled. Genes are tightly packed, with almost no non-coding DNA outside of D loop.

Differentiate between Computational and Experimental RNomics? 5 Marks

Computational RNomics

- Searching conserved intronic sequences by comparative analysis of introns
- Searching conserved intergenic sequences

Experimental RNomics

- PAGE separation of non-coding RNAs and sequencing
- Non-coding RNA enriched cDNA libraries and sequencing

Write the name of four database of Ensembles? 2 Marks

1. Ensemble Core Database
2. Ensemble EST Database
3. Ensemble Compara Database
4. Ensemble Variation Database

Why computational tool needed for genome sequencing? 2

Computational tools are used to align the sequenced DNA to produce physical maps of the genomes.

Explain the transposable elements? 5 Marks

Transposable Elements (TEs)

- Large portion of eukaryotic genome
- Thought to play important role in evolution of these genomes
- Move (Jump) from one location to another faster than chromosome replicate

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- Have a potential to increase in number
- Make up a large proportion of eukaryotic genome
- Detectable until blend into genome due to mutations

Promoter sequence difference between prokaryotes and eukaryotes?

- Promoter region (binding site), usually near coding region
- Binding can block (inhibit) expression

Exportin 5?

Exportin 5 is a member of the karyopherin nucleocytoplasmic transport factors. Argonautes are RNase H enzymes.

Define genome and genomics?

The complete set of DNA found in each cell is known as the genome and study of genome is called as genomics.

Genetic mapping and physical mapping? 3 Marks

Genetic Mapping

Genetic Mapping approximate locations of genes, relative to the locations of other genes, based on the rate of recombination.

Physical Mapping

Physical mapping is based on the direct analysis of DNA. Physical mapping places genes on the genomes in relation to distances measured in bp, kbp, and mbp.

Steps involve in Functional genomics? 3 Marks

- Subtracted cDNA libraries
- Differential display
- Representational difference analysis
- Suppression subtractive hybridization
- CDNA Microarrays
- Serial analysis of gene expression
- 2-D Gel electrophoresis

Types of DNA microarray?

Microarray Chips

- There are two types of DNA Chips/Microarrays:
- cDNA based microarray
- Oligonucleotide based microarray

cDNA Based Microarray Chips

- Chips are prepared by using cDNA- cDNA chips or cDNA microarray.

- The cDNAs are amplified. Then these immobilized on a solid support made up of nylon filter of glass slide
- The probe DNA are loaded by capillary action.
- Small volume of this DNA is spotted on solid surface
- DNA is delivered mechanically or in a robotic manner.
- When one DNA spotting is done, the pin is washed and loaded with fresh DNA to start the second cycle

DNA Based Microarray Chips

- Almost similar type procedure
- Gene expression profiling
- Discovery of drugs
- Diagnostics and genetic engineering
- Alternative splicing detection
- Functional genomics
- DNA sequencing
- Toxicological research (Toxicogenomics)

Human genome sequence challenges?

- Challenges to Sequence Human Genome
- Size
- Polymorphism
- Repeats -smaller repeats are technically difficult to sequence
- Some DNA sequences are repeated all over the genome

Mitochondrial genome?

Multiple identical circular chromosomes

~15-16 Kb in animals

200 kb to 2,500 kb in plants

Eukaryotic genomes?

Eukaryotic genomes are full of simple repeats, numerous types of transposable elements and other sequences.

Which genes are in Arabidopsis genomes that are not seen in other Eukaryotes organisms?

Arabidopsis genome contains genes encoding RNA polymerase subunits not seen in other Eukaryotic organisms.

What are four types of Transposable Elements?

Structural Genomics- Transposable Elements

Vast majority of TEs can be classified into four families:

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- D LINEs (Long Interspersed Nuclear Elements, autonomous)
- D SINEs (Short Interspersed Nuclear Elements, use LINE proteins for life cycle)
- D LTR elements (Long Terminal Repeats; derived from retroviruses)
- D DNA transposons (replicate without RNA intermediary)

What is Cytogenetic Mapping and what is meant by regions?

Cytogenetic Mapping

Microscopic examination of stained chromosome reveals a banding pattern. Chromosomal banding pattern. These methods allow a rough determination of locations, but not to yield a direct measure of distance.

Regions – designated based on their chromosomal position relative to the centromere.

Name of at least five model organisms? 5 Marks

1. E. coli- bacteria
2. S. cerevisiae- yeast
3. C. elegans- worm
4. D. melanogaster- fly
5. Danio rerio- zebrafish

How Xist activity related?

Xist activity is regulated by another 40,000 nt long ncRNA – Tsix, which contains anti-sense sequence of Xist and therefore is able to regulate Xist activity by base-pairing to it

Differentiate between cloning and hybridization?

Cloning

Genomes digested with restriction enzymes and inserted in vectors to produce genomic libraries.

- BACS
- YACS

Hybridization

To arrange large contigs of genomes to produce genetic maps and physical maps
Of genomes.

Different types of Genome mapping?

- Genetic mapping
- Physical mapping

Genetic mapping is based on the use of genetic techniques to construct maps showing the positions of genes and other sequence features on a genome.

Enlist three organizations that have joint project with Ensemble?

1. EMBL
2. EBI
3. WTSI

In what ways does genome sequence information help in research? (5)

Genome sequence information is useful in research by helping in identifying genes, their functions, and their arrangement in the genome. It also helps in comparative genomics, precision medicine, and biotechnology. Genome sequence data can provide insights into the molecular basis of various traits and can help in developing targeted therapies or personalized treatment plans. Combining genome sequence data with other experimental techniques can accelerate progress in various fields.

Genome function related? (3)

DNA transduction & it's stages? (5)

DNA transduction is a process by which foreign DNA is introduced into a bacterial cell by a bacteriophage virus. The stages of DNA transduction include:

- **Attachment**: The bacteriophage virus attaches to a specific receptor on the surface of the bacterial cell.
- **Penetration**: The viral DNA is then injected into the bacterial cell through the bacterial cell wall and membrane.
- **Replication**: Once inside the bacterial cell, the viral DNA replicates and synthesizes new viral proteins.
- **Packaging**: The viral DNA and proteins are then assembled into new virus particles.
- **Lysis**: Finally, the bacterial cell is lysed or ruptured, and the newly assembled virus particles are released into the surrounding environment.

Write the names of two protocols of clone independent sequencing?

The two protocols of clone independent sequencing are:

1. **Whole-genome shotgun sequencing**: This protocol involves randomly breaking the genomic DNA into small fragments, sequencing the fragments, and assembling them using computational algorithms.
2. **Hierarchical shotgun sequencing**: This protocol involves first sequencing larger fragments of genomic DNA to create a physical map, which is then used to guide the sequencing of smaller, overlapping fragments that are assembled into the final sequence.